

ESKDALEMUIR, United Kingdom

WMO: 031620

Lat: 55.312N

Lon: 3.207W

Elev: 794

StdP: 14.28

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.6	24.1	17.7	13.8	24.5	20.5	15.8	27.3	30.4	43.3	27.6	43.9	2.0	340	0.661

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	73.1	62.0	69.1	60.0	65.9	58.6	63.8	70.2	61.7	66.8	60.0	64.1	8.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.2	83.0	65.7	59.6	78.3	63.3	58.2	74.5	61.7	29.4	70.3	27.9	67.0	26.7	64.1	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.1	21.9	19.2	DB	15.4	78.8	4.2	3.5	12.4	81.3	9.9	83.3	7.6	85.3	4.5	87.9
			WB	14.9	67.0	4.1	2.0	11.9	68.4	9.5	69.6	7.2	70.7	4.2	72.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	45.7	36.8	37.1	39.3	43.3	48.3	53.2	56.5	56.0	52.2	46.9	41.1	37.0	(6)
(7)		DBStd	8.82	5.30	5.00	4.97	4.81	5.43	4.39	4.28	4.24	4.19	5.21	5.51	6.56	(7)
(8)		HDD50	2248	411	362	332	206	98	16	1	3	24	121	271	403	(8)
(9)		HDD65	7053	876	782	796	650	518	355	265	282	384	560	718	867	(9)
(10)		CDD50	672	0	0	0	6	45	111	204	188	90	26	2	0	(10)
(11)		CDD65	5	0	0	0	0	0	0	3	2	0	0	0	0	(11)
(12)		CDH74	76	0	0	0	0	3	13	38	22	0	0	0	0	(12)
(13)		CDH80	4	0	0	0	0	0	1	2	1	0	0	0	0	(13)
(14)	Wind	WSAvg	8.2	9.0	9.4	9.0	8.4	8.6	8.0	7.0	6.8	7.4	8.0	8.0	8.4	(14)
(15)	Precipitation	PrecAvg	65.4	7.3	5.3	5.4	3.7	3.9	3.9	4.4	5.2	5.5	6.8	6.7	7.2	(15)
(16)		PrecMax	96.5	14.5	15.8	11.1	6.9	10.2	9.7	10.6	15.5	10.7	13.9	17.0	22.4	(16)
(17)		PrecMin	40.5	1.5	0.7	1.8	0.7	0.2	0.5	1.1	0.9	0.6	1.9	1.3	1.8	(17)
(18)		PrecStd	10.9	3.4	3.2	2.3	1.5	2.1	1.7	2.0	3.0	2.3	3.3	3.0	3.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.1	51.0	57.9	66.0	73.1	75.3	78.5	77.3	69.4	61.2	55.6	51.9	(19)
(20)			MCWB	48.5	47.4	48.0	53.8	60.3	61.9	65.0	65.0	60.5	56.0	54.3	50.7	(20)
(21)		2%	DB	48.0	48.5	51.6	59.1	67.3	69.9	73.5	70.6	64.5	58.0	52.8	49.8	(21)
(22)			MCWB	46.8	46.8	46.7	50.1	56.8	59.9	62.6	62.0	57.6	55.4	51.4	48.8	(22)
(23)		5%	DB	46.2	46.6	49.0	54.6	62.5	65.6	69.1	66.2	61.7	56.4	51.2	48.3	(23)
(24)			MCWB	45.0	45.1	45.6	48.1	54.0	57.7	60.8	60.0	56.6	54.2	49.9	47.2	(24)
(25)		10%	DB	44.8	45.0	47.0	51.4	58.2	61.9	65.5	63.3	59.6	54.7	49.6	46.5	(25)
(26)			MCWB	43.7	43.4	44.4	46.8	51.4	55.7	59.2	58.3	55.8	52.7	48.3	45.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.1	49.1	50.8	54.4	61.8	64.1	66.6	66.2	62.6	57.9	54.3	51.2	(27)
(28)			MCD B	49.9	49.9	55.1	63.1	70.4	71.1	75.1	73.6	66.8	59.4	55.2	51.9	(28)
(29)		2%	WB	47.0	47.1	48.2	51.3	57.7	61.0	64.1	63.3	59.8	56.3	51.9	49.2	(29)
(30)			MCD B	47.6	48.1	50.5	56.4	65.2	68.1	71.1	68.6	62.1	57.4	52.5	49.7	(30)
(31)		5%	WB	45.4	45.5	46.3	49.4	55.1	58.6	61.8	61.1	58.0	54.7	50.2	47.4	(31)
(32)			MCD B	46.1	46.5	48.1	53.1	60.8	64.0	67.2	64.8	60.2	55.9	50.9	48.1	(32)
(33)		10%	WB	43.9	43.8	44.9	47.7	52.6	56.6	60.0	59.5	56.5	53.0	48.4	45.5	(33)
(34)			MCD B	44.7	44.8	46.7	50.6	57.1	60.6	64.3	62.2	58.9	54.4	49.4	46.4	(34)
(35)	Mean Daily Temperature Range	MDBR		8.3	10.0	12.0	14.5	15.9	14.7	15.0	14.3	13.7	11.8	10.0	9.3	(35)
(36)		5% DB	MCD BR	9.3	11.1	15.3	20.8	23.7	22.2	22.2	19.0	17.5	12.1	10.3	9.9	(36)
(37)			MCWBR	9.3	9.5	10.8	13.2	14.1	13.6	13.0	11.6	11.4	9.5	9.8	9.9	(37)
(38)		5% WB	MCD BR	9.1	10.2	12.8	17.0	20.2	19.2	19.1	16.3	13.6	10.7	10.0	9.9	(38)
(39)			MCWBR	9.3	9.4	10.1	11.6	12.7	12.7	11.9	10.7	10.2	9.5	9.9	10.0	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.314	0.328	0.355	0.374	0.376	0.374	0.373	0.362	0.348	0.333	0.308	0.299	(40)
(41)		taud		2.252	2.347	2.305	2.282	2.321	2.352	2.354	2.399	2.444	2.432	2.334	2.225	(41)
(42)		Ebn at Noon		189	237	257	268	275	276	274	270	258	233	193	165	(42)
(43)		Edh at Noon		18	24	31	36	37	36	36	32	27	22	16	15	(43)
(44)	All-Sky Solar Radiation	RadAvg		174	377	709	1145	1495	1458	1406	1137	816	449	223	129	(44)
(45)		RadStd		25	38	73	124	114	144	156	90	68	54	21	11	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(46)
(47)	Regional (55 neighbors)	(m)									(47)

Nomenclature: See separate page